|INCH-POUND|

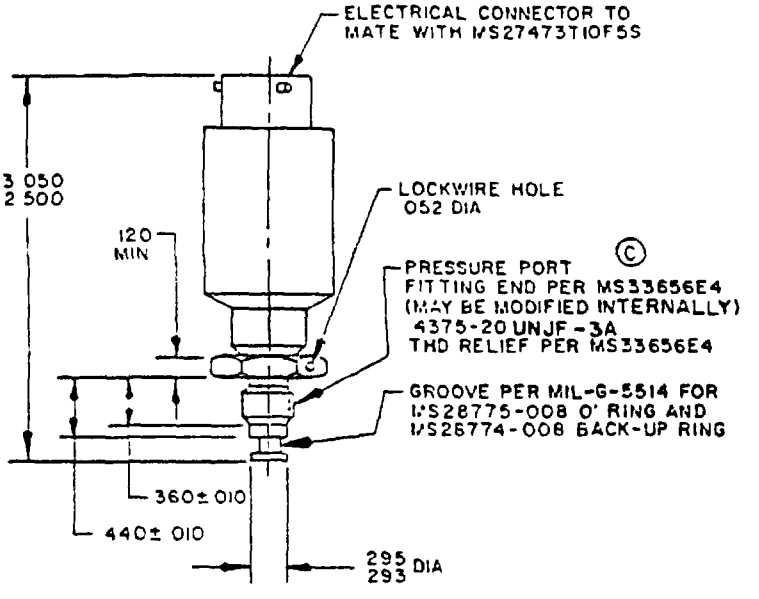
MIL-S-9395/31C
30 October 1989
SUPERSEDING
MIL-S-9395/31B
23 July 1980

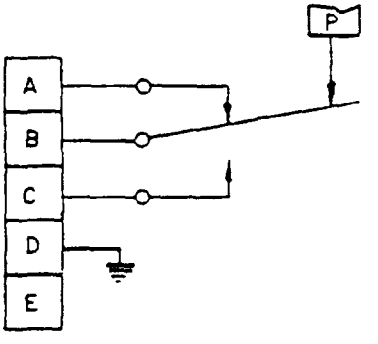
MILITARY SPECIFICATION SHEET

SWITCHES, PRESSURE, (GAGE), TYPE II, 5 AMPERES

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation MIL-S-9395.





SCHEMATIC

Inches	mm	Inches	mm
.010	0.25	.4375	11.112
.052	1.32	.440	11.18
.120	3.05	.750	19.05
.293	7.44	.866	21.99
.295	7.49	2.500	63.50
.360	9.14	3.050	77.47

WEIGHT: NOT TO EXCEED .2 lbs.

FIGURE 1. Switches.

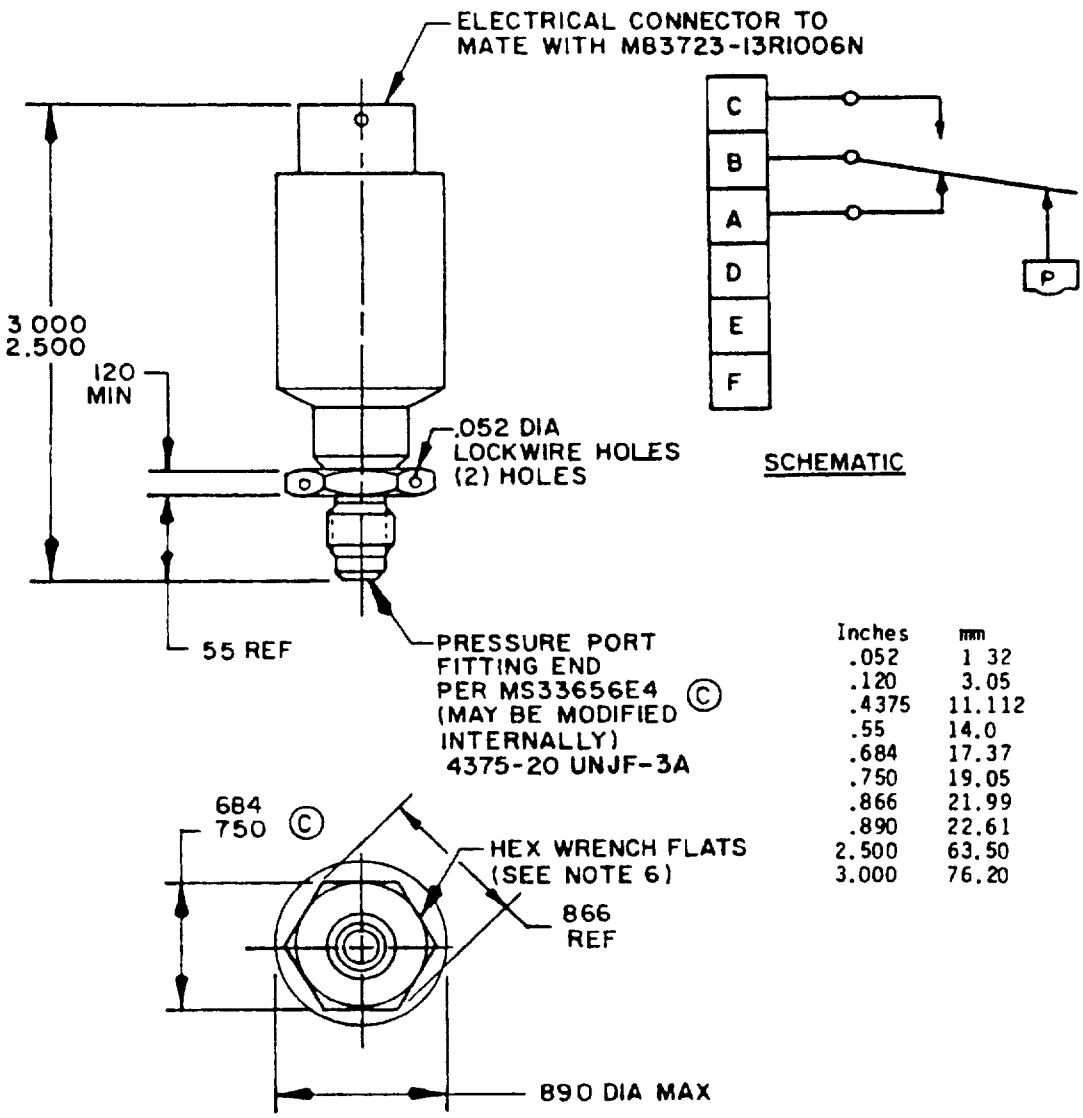
Configuration 1

© denotes changes

AMSC N/A FSC 5930

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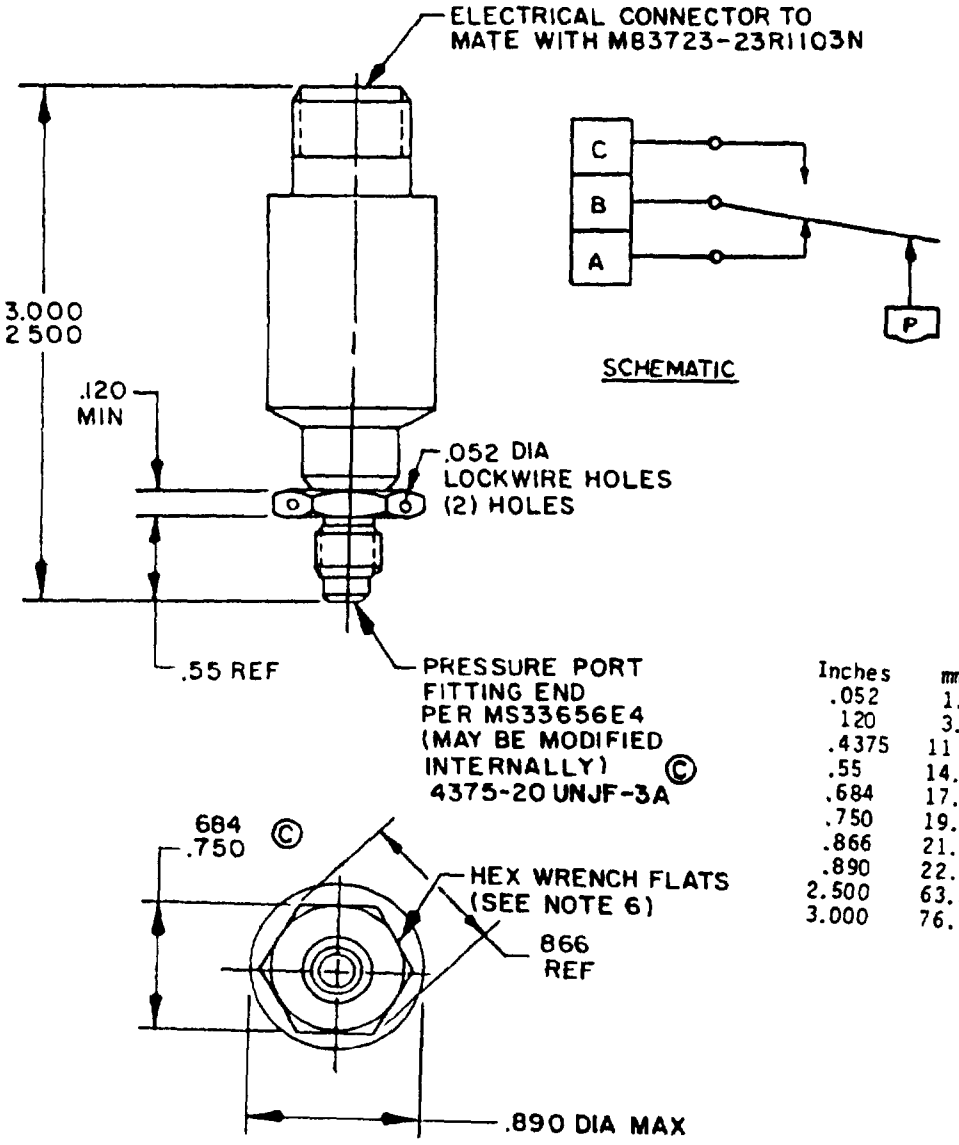
Inches	mm
.052	1 32
.120	3.05
.4375	11.112
.55	14.0
.684	17.37
.750	19.05
.866	21.99
.890	22.61
2.500	63.50
3.000	76.20

WEIGHT: NOT TO EXCEED 2 lbs

Configuration 2

FIGURE 1 Switches - Continued.

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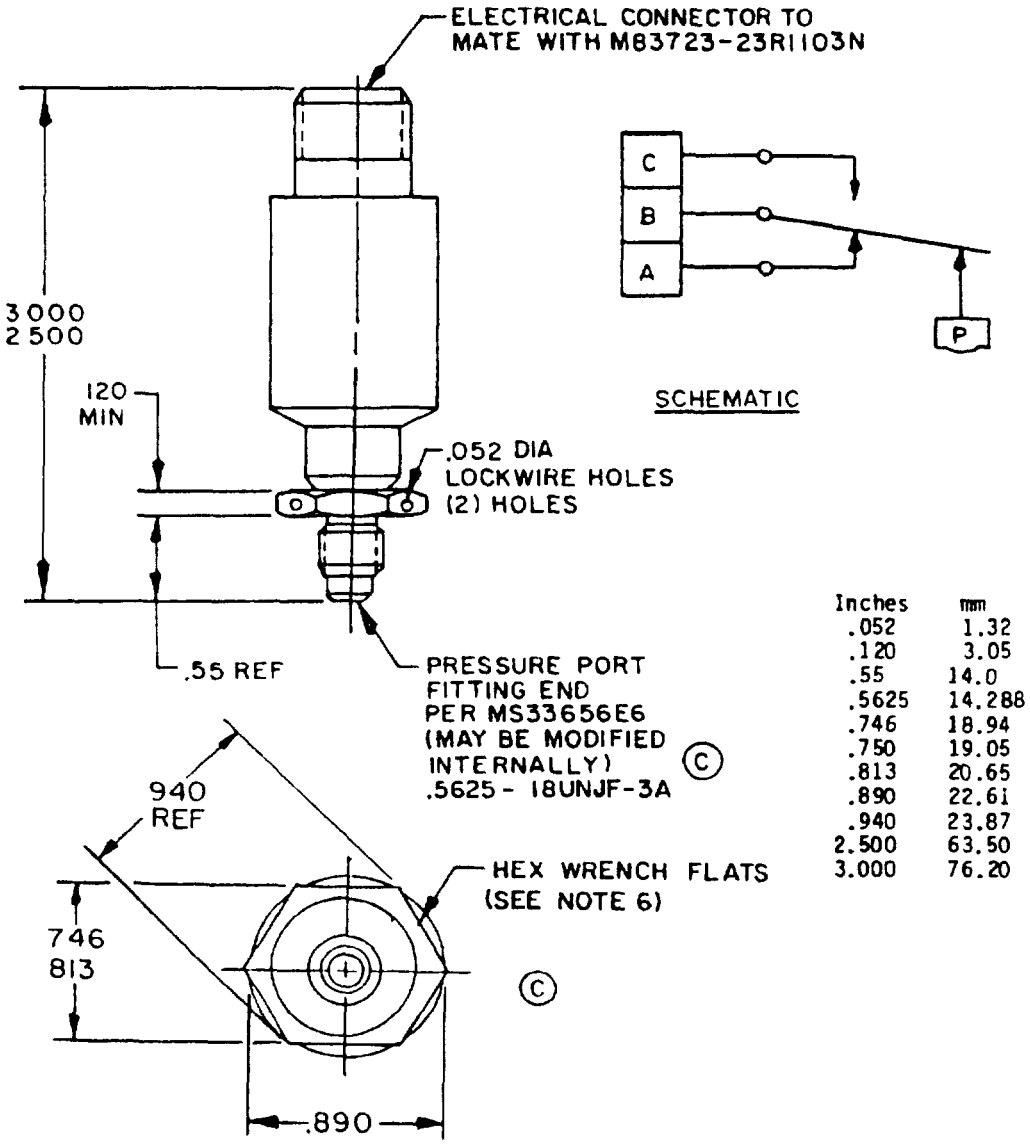
Inches	mm
.052	1.32
120	3.05
.4375	11 112
.55	14.0
.684	17.37
.750	19.05
.866	21.99
.890	22.61
2.500	63.50
3.000	76.20

WEIGHT: NOT TO EXCEED .2 lbs.

Configuration 3

FIGURE 1. Switches - Continued

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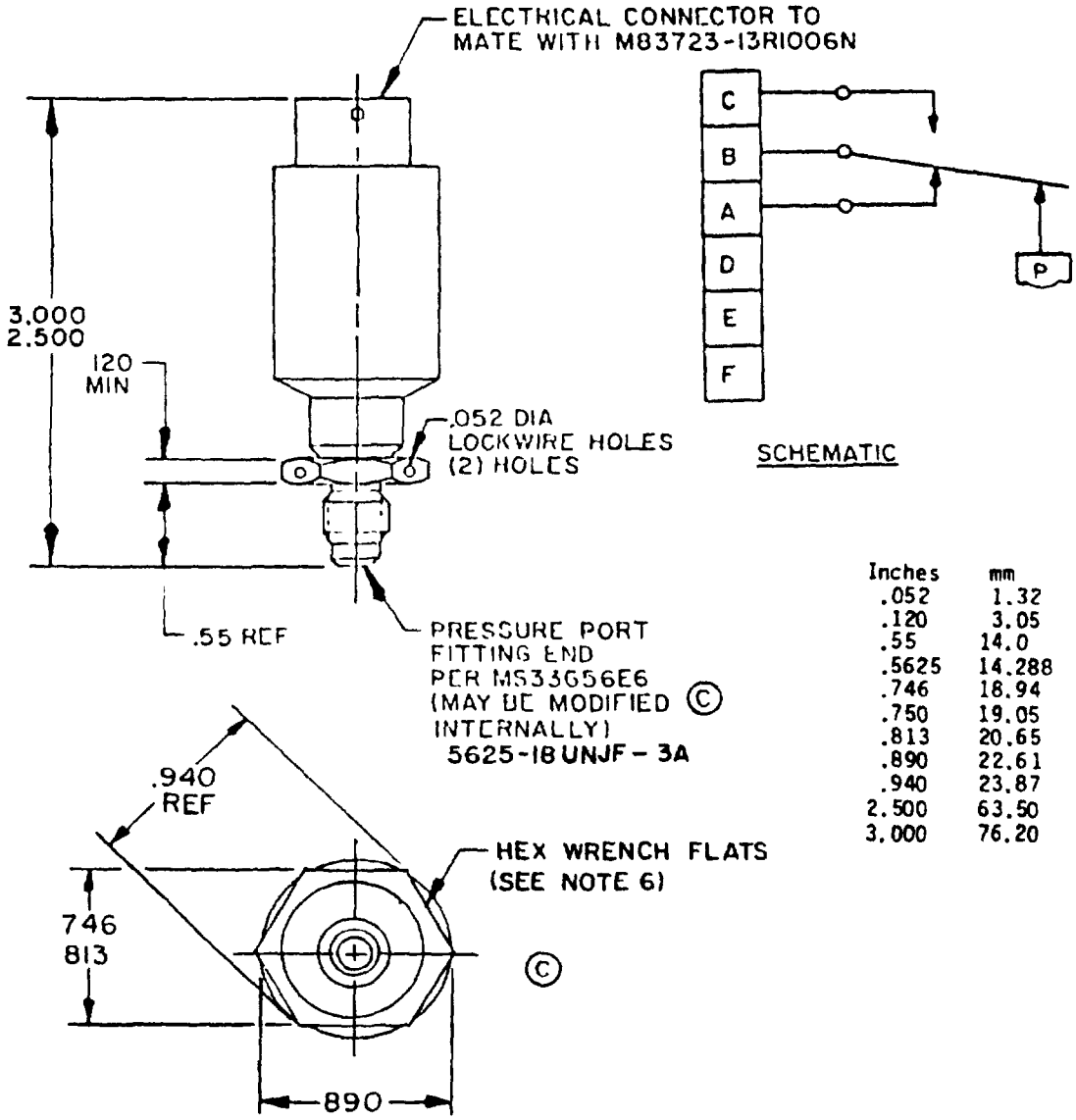
Inches	mm
.052	1.32
.120	3.05
.55	14.0
.5625	14.288
.746	18.94
.750	19.05
.813	20.65
.890	22.61
.940	23.87
2.500	63.50
3.000	76.20

WEIGHT: NOT TO EXCEED .2 lbs.

Configuration 4

FIGURE 1. Switches - Continued.

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Inches	mm
.052	1.32
.120	3.05
.55	14.0
.5625	14.288
.746	18.94
.750	19.05
.813	20.65
.890	22.61
.940	23.87
2.500	63.50
3.000	76.20

WEIGHT. NOT TO EXCEED .2 lbs.

Configuration 5

NOTES
1. Dimensions are in inches
2. Metric equivalents are given for general information only.
3. Exact shape of switch is optional provided outline dimensions specified are not exceeded and mounting holes and connector locations are as specified.
4. Schematics shown are for switches with pressure ports exposed to room ambient.
5. Unless otherwise specified, tolerance is ±.005 (0.13 mm).
6 A minimum of two wrench flats is needed.

FIGURE 1. Switches - Continued.

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REQUIREMENTS:

Dimensions, weight, and electrical schematic: See figure 1

Materials: All external parts including pressure port, housing, and electrical connector shall be fabricated from corrosion-resistant (stainless) steel suitable for welding External parts shall be joined by welding; solder joints shall not be allowed

Calibration. See tables I, II, III, and IV.

Proof pressure: 4500 lb_f/in^2. The electrical chamber shall withstand 4500 lb_f/in^2 without rupture

System pressure. 3000 lb_f/in^2.

Burst pressure: 7500 lb_f/in^2.

Electrical ratings.

Operating voltage. 28 V dc.

Current rating 5 amperes resistive
2 amperes inductive

Seal

Pressure chamber Media proof Subject switches to proof pressure for 2 minutes using hydraulic fluid in accordance with MIL-H-5606 with chamber pressure continuously being monitored. Isolate the chamber at proof pressure, with the chamber disconnected from the pressure source. Under that condition, the pressure shall not drop more than 5 lb_f/in^2 for the first 30 seconds to allow for stabilization of test equipment. No pressure loss is allowed thereafter for the remainder of the 2 minutes

Electrical chamber See table I.

Electrical connector. See figure 1.

Pressure port See figure 1.

Media. Hydraulic fluid in accordance with MIL-H-5606 or MIL-H-83282, fuel in accordance with MIL-T-5624; lubricating oil in accordance with MIL-L-7808.

High temperature (operating and nonoperating) B (+275°F).

Low temperature (operating and nonoperating) D (-65°F).

Altitude: C (except 80,000 feet)

Shock: C (100 g).

Vibration: S (test condition D, method 204 of MIL-STD-202).

Life (mechanical) A (100,000 cycles)

Life (electrical). C (50,000 cycles)

Acceleration C (8 g).

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Pulsation amplitude: E (10 percent)

Pulsation frequency: D (500 ±50 Hz).

Pressure rise: F (500,000 $lb_f/in^2/s$).

Dielectric withstanding voltage (at reduced barometric pressure). Applicable.

Connector torque. Applicable.

Ⓒ Part or Identifying Number (PIN). Consists of the prefix "M9395/31-" followed by a five-character code. The code identifies the configuration, seal, and pressure setting mode (code from table I), high-pressure setting to within 100 lb_f/in^2 (code from table II) followed by high-pressure setting to within 25 lb_f/in^2 and applicable tolerance (code from table III); and low-pressure setting to within 100 lb_f/in^2 (code from table II) followed by low-pressure setting to within 25 lb_f/in^2 and applicable tolerance (code from table III) When a pressure differential is used for one of the actuation values, two codes from table IV are selected to define the differential spread

	Example			
	M9395/31-	V	DB	FH
Prefix.				
Configuration, seal, and pressure setting mode Code "V" of table I.				
High-pressure setting (400 lb_f/in^2) and (±25, ±25 lb_f/in^2). Code "D" of table II Code "B" of table III				
Low-pressure setting differential. (125 lb_f/in^2) and (+175 lb_f/in^2). Code "F" of table IV Code "H" of table IV				

PIN M9395/31-VDBFH describes a switch of configuration "4" which actuates on increasing pressure at 400 ±25 lbf/in^2 and with a deadband value of 125 to 175 lbf/in^2.

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TABLE I. Codes for combinations of configurations and pressure setting modes

	Configuration 1		Configuration 2		Configuration 3		Pressure setting mode	
	Electrical chamber		Electrical chamber		Electrical chamber		High setting	Low setting
	Hermetic	Unsealed	Hermetic	Unsealed	Hermetic	Unsealed		
Code ---	A	D	G	K	N	R	At (or max) 1/	At (or min) 1/
Code ---	B	E	H	L	P	S	At (or max) $\overline{1}$/	Differential $\overline{2}$/
Code ---	C	F	J	M	Q	T	Differential $\overline{2}$/	At (or min) $\overline{1}$/

	Configuration 4		Configuration 5		Pressure setting mode	
	Electrical chamber		Electrical chamber		High setting	Low setting
	Hermetic	Unsealed	Hermetic	Unsealed		
Code ---	U	X	1	4	At (or max) 1/	At (or min) 1/
Code ---	V	Y	2	5	At (or max) $\overline{1}$/	Differential $\overline{2}$/
Code ---	W	Z	3	6	Differential $\overline{2}$/	At (or min) $\overline{1}$/

1/ Setting values are designated by characters from tables II and III
$\overline{2}$/ Setting values are designated by characters from table IV.

TABLE II Codes for pressure settings to within 100 lb_f/in^2.

Code	Pressure (lb_f/in^2)	Code	Pressure (lb_f/in^2)	Code	Pressure (lb_f/in^2)	Code	Pressure (lb_f/in^2)
A---	100	J---	900	S---	1700	1---	2500
B---	200	K---	1000	T---	1800	2---	2600
C---	300	L---	1100	U---	1900	3---	2700
D---	400	M---	1200	V---	2000	4---	2800
E---	500	N---	1300	W---	2100	5---	2900
F---	600	P---	1400	X---	2200	6---	3000
G---	700	Q---	1500	Y---	2300		
H---	800	R---	1600	Z---	2400		

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TABLE III Codes for pressure settings to within 25 lb_f/in^2 and tolerances

	Unit				Tolerance (lb_f/in^2)
	0	+25	+50	+75	
Code ---	A	B	C	D	±25
Code ---	E	F	G	H	±50
Code ---	J	K	L	M	±75
Code ---	N	P	Q	R	±100
Code ---	S	T	U	V	±150
Code ---	W	X	Y	Z	±200
Code ---	1	2	3	4	±300
Code ---	5	6	7	8	±400
Code ---	9	0	+	-	Min or Max

TABLE IV Codes for differential pressure settings.

Code	Differential value (lb_f/in^2)	Code	Differential value (lb_f/in^2)
A ---	0	M ---	275
B ---	25	N ---	300
C ---	50	P ---	325
D ---	75	Q ---	350
E ---	100	R ---	375
F ---	125	S ---	400
G ---	150	T ---	425
H ---	175	U ---	450
J ---	200	V ---	475
K ---	225	W ---	500
L ---	250		

QUALIFICATION

Single submission Restricted to switch submitted.
Group submission See table V

TABLE V. Extent of qualification.

PIN	Number of samples required	Tests	Qualifies
M9395/31-AADAA	2 each resistive	Complete in accordance with qualification inspection of MIL-S-9395	All configurations
M9395/31-DADAA	2 each resistive		
M9395/31-G4ZXV	2 each inductive		
M9395/31-N4ZXV	2 each inductive		

NOTE.
Design limitations (actuation values and tolerances, deadband and deactuation values and tolerances) should be coordinated with manufacturer(s) listed on the QPL for this specification sheet before specifying a particular "M" number The fact that operating characteristics can be coded does not necessarily mean that it can be manufactured or procured

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CONCLUDING MATERIAL

Custodians
Army - ER
Navy - EC
Air Force - 85

Review activities
Army - AV
Air Force - 11, 99
DLA - ES

User activities
Army - AT
Navy - AS, MC, SH
Air Force - 19

Preparing activity.
Air Force - 85

Agent
DLA - ES

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